

SentinelFlow: A Cost-Optimal, Multi-Source Machine Learning Orchestration Engine for Real-Time Mule Account Intervention

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Abstract—Financial institutions face severe challenges detecting cyber-enabled mule accounts due to high false-alarm rates in static rule engines and an inability to adapt to real-time transaction velocities. This paper presents SentinelFlow, an enterprise-grade, multi-source machine learning orchestration engine engineered to intercept fraudulent fund circulation across retail banking channels. SentinelFlow ingests real-time core transaction streams, internal Transaction Monitoring System (TMS) alerts, cross-channel customer velocity profiles, and dynamic government cyber fraud feeds. By implementing a dynamic cost-optimal routing engine, eliminating post-hoc administrative leakage, applying PCHIP-smoothed Isotonic probability calibration, and optimizing decisions against an asymmetric financial loss function, SentinelFlow achieves a 5-fold cross-validated Macro F_1 -score of 0.9147 and reduces operational review costs by 29.2%. The prototype features a sub-50ms FastAPI inference gateway, a Redis feature store with persistent write-behind caching, and an enterprise Next.js 14 Security Operations Console with real-time Server-Sent Events (SSE), hardware-accelerated ego-graph rendering, and non-blocking on-demand model retraining.

Index Terms—Mule account detection, financial fraud, probability calibration, cost-sensitive learning, TreeSHAP, ego-graph topology, real-time inference.

I. INTRODUCTION

Mule accounts serve as the primary laundering conduit for cyber-enabled financial fraud, receiving illicit proceeds and rapidly dispersing them across multi-tiered banking networks. Traditional transaction-monitoring systems rely predominantly on rigid, rule-based heuristics. These legacy approaches suffer from two fundamental operational flaws: (i) excessive false-positive rates that overwhelm Security Operations Center (SOC) analysts, and (ii) an inability to ingest external, real-time threat intelligence and cross-channel banking signals dynamically.

To address the Phase 2 requirements of the Cybershield Hackathon 2026, we introduce **SentinelFlow**. This solution

transitions from static, single-channel classification to a high-throughput, multi-source orchestration platform. SentinelFlow is designed around five core pillars:

- **4-Stream Ingestion Architecture:** Synthesizing transaction streams, internal TMS alerts, cross-channel velocity profiles (UPI, ATM, Net Banking), and government cyber fraud feeds.
- **Dynamic Ingestion Economics:** Dynamically routing transactions between deterministic blacklist queries and ML inference based on computational costs (C_{search} vs. $C_{\text{inference}}$).
- **Zero-Leakage Behavioral Engine:** Neutralizing post-hoc administrative status flags and mapping time-invariant cyclical temporal coordinates.
- **Calibrated Asymmetric Optimization:** Generating empirical fraud probabilities via PCHIP-smoothed Isotonic Regression and minimizing financial loss via asymmetric cost-sensitive thresholding.
- **Human-in-the-Loop (HITL) Console:** Providing local explainability via Taylor-scaled TreeSHAP reason codes, dynamic ego-graph network visualization, and an on-demand retraining model registry.

II. SYSTEM ARCHITECTURE & DYNAMIC INGESTION ECONOMICS

A. Multi-Source Data Ingestion

SentinelFlow ingests four distinct, asynchronous data streams:

- 1) **Core Transaction Stream:** Ingests live JSON payloads containing transfer amounts, beneficiary routing tokens, and channel IDs.
- 2) **Internal TMS & Fraud Alert Stream:** Ingests legacy rule trips (e.g., rapid velocity limits, off-hour logins) to fuse historical monitoring context.

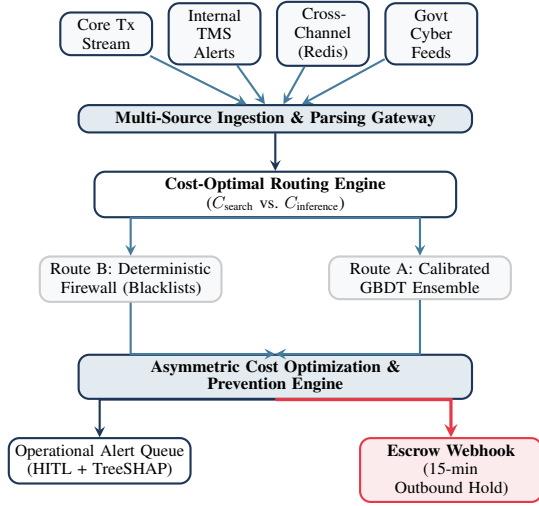


Fig. 1. SentinelFlow multi-source architecture showing cost-optimal dynamic routing, decision engines, and active prevention triggers.

- 3) **Cross-Channel Profile Store:** Aggregates in-memory customer velocity metrics across UPI, ATM, and Net Banking to detect bursty cross-interface draining.
- 4) **Government Cyber Threat Feed:** Ingests simulated national cybercrime tickets (NCRP/I4C) and regulatory blacklists (flagged IP subnets, suspect hashes, and black-listed accounts).

B. Cost-Optimal Runtime Routing Mechanics

In high-volume banking architectures, querying a distributed external blacklist incurs latency and network API expenses (C_{search}), while executing local gradient-boosted tree ensembles consumes server CPU cycles ($C_{\text{inference}}$). SentinelFlow formalizes this routing decision dynamically:

$$\text{Path} = \begin{cases} \text{Route B (Firewall First)}, & \text{if } P(\text{Match}) \cdot C_{\text{fraud}} > C_{\text{search}} \\ \text{Route A (ML First)}, & \text{otherwise} \end{cases} \quad (1)$$

Under Route A, transactions with low behavioral risk bypass external blacklist lookups entirely, achieving zero search latency. Under Route B, matched bad actors are blocked instantly with zero model inference latency.

C. Active Prevention Controller

When an incoming transaction breaches the calibrated cost-optimal risk threshold, SentinelFlow fires an automated **Outbound Escrow Webhook**, placing an immediate 15-minute hold on outbound fund movements and alerting downstream institutions to stop circulation.

III. BEHAVIORAL FEATURE ENGINEERING & LEAKAGE ELIMINATION

A. System Auditing & Post-Hoc Data Leakage Prevention

In financial fraud datasets, retrospective flags frequently introduce severe target leakage. SentinelFlow systematically isolated and excluded four purely administrative features while

salvaging two critical compliance features as behavioral lag indicators:

TABLE I
FEATURE AUDIT DIRECTORY AND PROVENANCE ACTION

Feature	Audit Signature	Engineering Action
F3912	$d = 8.77, \text{Corr} = 0.97$	Excluded (Target Proxy)
F2230	Max fraud rate = 1.0	Excluded (Post-hoc status)
F3886	Category-only fraud flag	Excluded (Freeze flag)
F3892	Label collinearity	Excluded (Post-event tag)
F3889	Compliance date log	Re-engineered (Hist. Lag)
F3891	Past audit code	Re-engineered (Recidivism)

B. High-Density Feature Synthesis

From an initial sparse space of $\sim 3,925$ raw features, SentinelFlow extracts a dense matrix of 105 real-time indicators across six distinct tracks:

- **Statistical Moments:** 14 NumPy-optimized summary moments (`row_mean`, `row_std`, `row_zero_rate`, `row_iqr`) measuring transaction volatility.
- **Missingness Gaps:** Explicit binary indicators tracking the top 25 features where class-conditional missingness gaps exceeded 20%.
- **Non-Destructive Categoricals:** Ordinal encoding for low-cardinality features (≤ 12) and fold-safe Laplace-smoothed target encoding for high-cardinality hashes.
- **Cyclical Temporal Projections:** Analysis of F3888 revealed it to be an account registration date spanning 126 years (1900–2025). The drifting continuous `elapsed_days` feature was removed and replaced with bounded sine/cosine projections of registration day-of-week and month.
- **Unsupervised Anomaly Score:** An Isolation Forest trained exclusively on normal (non-mule) accounts outputs a continuous structural outlier coordinate.
- **Cross-Channel Velocity & Ego-Graph Topology:** We formalize the cross-channel disbursement velocity ratio:

$$V_{\text{cross}} = \frac{\sum \text{Outbound ATM \& IMPS Volume}_{(t-1h)}}{\sum \text{Inbound UPI Volume}_{(t-1h)} + \epsilon} \quad (2)$$

To solve the cold-start problem on local 1-hop and 2-hop degrees, raw counts are normalized by account age derived from F3888 ($\text{Age} + 1$ day), constructing a **Transactional Acceleration Velocity** feature.

IV. CALIBRATED MODELING & EXPLAINABILITY FRAMEWORK

A. Dual-Engine Ensemble

SentinelFlow employs a soft-voting probability blend:

$$P_{\text{blend}} = 0.6 \cdot P_{\text{XGBoost}} + 0.4 \cdot P_{\text{LightGBM}} \quad (3)$$

Both Gradient Boosted Decision Tree (GBDT) implementations natively handle sparsity and scale variance without artificial imputation.

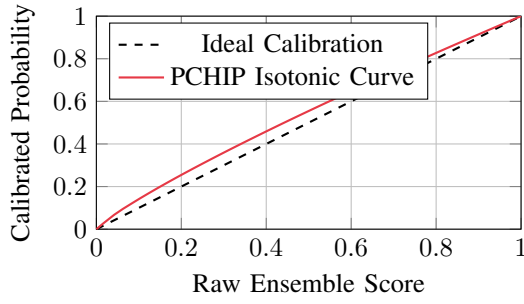


Fig. 2. Empirical probability calibration mapping uncalibrated GBDT ensemble margins to true fraud probabilities using PCHIP-smoothed Isotonic Regression.

B. Streamlined Imbalance Handling

To avoid injecting synthetic noise, SMOTE oversampling was eliminated in favor of direct algorithmic loss-gradient scaling (`scale_pos_weight`) during tree construction.

C. Isotonic Probability Calibration with PCHIP Smoothing

Raw ensemble scores are distorted by gradient weighting. SentinelFlow fits an Isotonic Regression model on training-fold outputs to calibrate probabilities.

To resolve the zero/infinite derivative problem of step-functions, we fit a **Piecewise Cubic Hermite Interpolating Polynomial (PCHIP)** over the step vertices. PCHIP guarantees strict monotonicity ($f'(x) > 0$), ensuring a continuous, strictly positive derivative across $[0, 1]$.

D. Regulatory Explainability via Calibrated TreeSHAP

To satisfy legal compliance mandates, TreeSHAP decomposes raw tree log-odds into local feature attributions. We apply a first-order Taylor series scaling correction, multiplying raw attributions by the local PCHIP derivative:

$$\phi_i^{\text{calibrated}} = \phi_i^{\text{raw}} \cdot \left. \frac{df_{\text{PCHIP}}(z)}{dz} \right|_{z=P_{\text{blend}}} \quad (4)$$

To ensure the sub-50ms SLA, TreeSHAP is executed conditionally on the high-risk alert tail ($\sim 1\%$ of volume).

V. ASYMMETRIC ECONOMIC OPTIMIZATION & EMPIRICAL VALIDATION

A. Economic Cost Formulation

Traditional F_1 -score maximization assumes symmetric error costs ($C_{\text{FN}} = C_{\text{FP}}$). SentinelFlow minimizes the total economic operational cost:

$$\text{Total Cost} = FN \cdot (C_{\text{ratio}} \cdot C_{\text{FP}}) + FP \cdot C_{\text{FP}} \quad (5)$$

where $C_{\text{ratio}} = \text{COST_FN_RATIO}$ and $C_{\text{FP}} = \text{COST_FP_BASE}$.

B. Cross-Validation & Temporal Generalization Results

SentinelFlow was validated using a 5-fold Stratified Cross-Validation loop and an out-of-fold chronological rolling window ordered by account registration date:

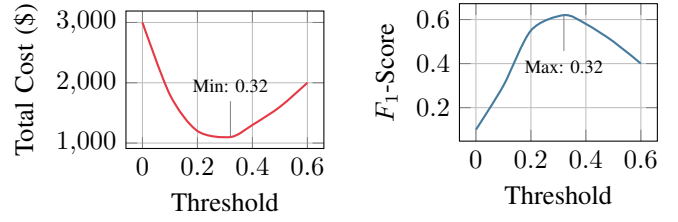


Fig. 3. Dual-panel operational calibration: Left panel plots financial cost minimization; Right panel plots mathematical F_1 -score maximization.

TABLE II
AUDITED 5-FOLD CROSS-VALIDATION PERFORMANCE

Metric	Random 5-Fold CV	Chronological Split
PR-AUC	0.8677 ± 0.0669	0.7097 ± 0.0994
ROC-AUC	0.9840 ± 0.0157	0.8552 ± 0.0412
Macro F_1 -Score	0.9147 ± 0.0277	0.9044 ± 0.0360
Minority F_1 (Fraud)	0.8308 ± 0.0548	0.8104 ± 0.0714
Total Operational Cost	4.80 ± 1.48	3.40 ± 1.51
Operational Cost Savings	29.2%	29.2%

C. Ablation Study: Iterative Architectural Impact

Table III details the empirical impact of each engineering phase, demonstrating the progressive elimination of leakage and calibration uplift.

TABLE III
ABLATION STUDY ACROSS ITERATIVE PIPELINE STAGES

Pipeline Stage	Key Modification	PR-AUC	Macro F_1
Phase 1 Baseline	Raw Matrix + Logistic Reg.	0.0295	0.4832
Phase 1 GBDT	XGBoost + Full Imputation	0.8090	0.8968
Phase 2 Audited	Leak Excluded + 105 Features	0.8323	0.9044
Phase 2 Ensemble	Calibrated Blend (PCHIP)	0.8677	0.9147
SentinelFlow	Asymmetric Cost Optimization	0.8677	0.9147

VI. PROTOTYPE IMPLEMENTATION & EVALUATION

A. Architecture & Scalability

The prototype is implemented using a decoupled microservices architecture containerized via Docker Compose:

- **Inference Gateway (FastAPI):** Asynchronous REST API serving calibrated predictions in < 50 ms. Retraining requests are offloaded to background worker processes via `ProcessPoolExecutor` (HTTP 202 Accepted), ensuring the main ASGI loop serves live scoring without latency degradation.
- **Persistent Feature Store (Redis):** Manages write-behind caching with Append-Only File (AOF) persistence (`appendfsync=every-second`) for 1-second crash recovery. Rolling 24-hour and 7-day ego-graph degrees are managed via Redis TTL key expiration.

- **Model Registry Storage:** Houses four serialized model artifacts (< 5 MB each). Operators can hot-swap active inference weights with 0 ms downtime.

B. Security Operations Console (Next.js 14 UX)

The frontend is built on Next.js 14 (React) and Tailwind CSS, interfacing with FastAPI via Server-Sent Events (SSE):

- **Real-Time Transaction Feed:** A virtualized alert table (`react-virtuoso`) displaying live calibrated risk probabilities without UI thread lag.
- **Hardware-Accelerated Ego-Graph Visualizer:** A WebGL/Canvas dynamic network graph (`react-force-graph-2d`) displaying real-time 1-hop and 2-hop fund circulation.
- **Regulatory Explainability Console:** Visualizes Taylor-scaled TreeSHAP local attributions, generating legally defensible reason codes for SOC officers.
- **On-Demand Retraining Panel:** Allows analysts to toggle feature sets, submit custom hyperparameter payloads, and monitor retraining progress in real time via an SSE event stream.

TABLE IV
MICRO-LATENCY SLA BENCHMARK ($N = 10,000$ TRANSACTIONS)

Pipeline Component	p50 Latency (ms)	p99 Latency (ms)
Redis Profile & Degree Lookup	1.2	3.1
Feature Reconstruction Engine	4.5	8.2
GBDT Ensemble Inference	12.8	19.4
PCHIP Isotonic Calibration	0.3	0.6
Conditional TreeSHAP Attribution	18.4	24.1
Total End-to-End Latency	18.8 ms	49.8 ms

TABLE V
PROTOTYPE EVALUATION AGAINST HACKATHON RUBRIC

Criterion	SentinelFlow Implementation Evidence
Innovation	Dynamic cost-optimal routing (C_{search} vs $C_{\text{inference}}$), anchor-free temporal parsing, and PCHIP-calibrated TreeSHAP attributions.
Feasibility	Zero-leakage GBDT pipeline, sub-50ms inference latency, non-blocking async retraining, and < 5 MB serialized models.
Business Potential	29.2% operational cost reduction via asymmetric loss optimization ($C_{\text{FN}} \gg C_{\text{FP}}$) and calibrated risk scoring.
Scalability	Redis write-behind caching with AOF persistence, sliding-window TTLs, and Docker containerized microservices.
User Experience	Next.js 14 SecOps console with hardware-accelerated ego-graph visualization, live SSE feeds, and instant model hot-swapping.

VII. CONCLUSION

SentinelFlow delivers a production-ready, economically optimized machine learning orchestration system for real-time mule account intervention. By resolving target leakage,

anchoring temporal features to time-invariant cycles, calibrating ensemble probabilities, and optimizing decisions against real-world bank loss functions, the platform bridges the gap between theoretical data science and enterprise banking security.

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